MEMORY FORENSICS REPORT

MFF v2 · Post-Volatility Analysis Framework

OVERALL SEVERITY: CRITICAL

Case ID	`case01_baseline_vs_case03_t1059_attack`
Generated	`2026-04-09 20:15:41 UTC`
Baseline capture	`case01_baseline`
Attack capture	`case03_t1059_attack`
Baseline path	`/MFF/cases/case01_baseline`
Attack path	`/MFF/cases/case03_t1059_attack`

1. EXECUTIVE SUMMARY

Metric	Value	Metric	Value
New Processes (Attack Only)	16	Critical Risk Processes	3
Gone Processes (Baseline Only)	16	ATT&CK Techniques Observed	2
New Network Connections	133	ATT&CK Tactics Observed	1
Flagged Connections	4	IOCs Extracted	56

ATT&CK; Tactics Observed:

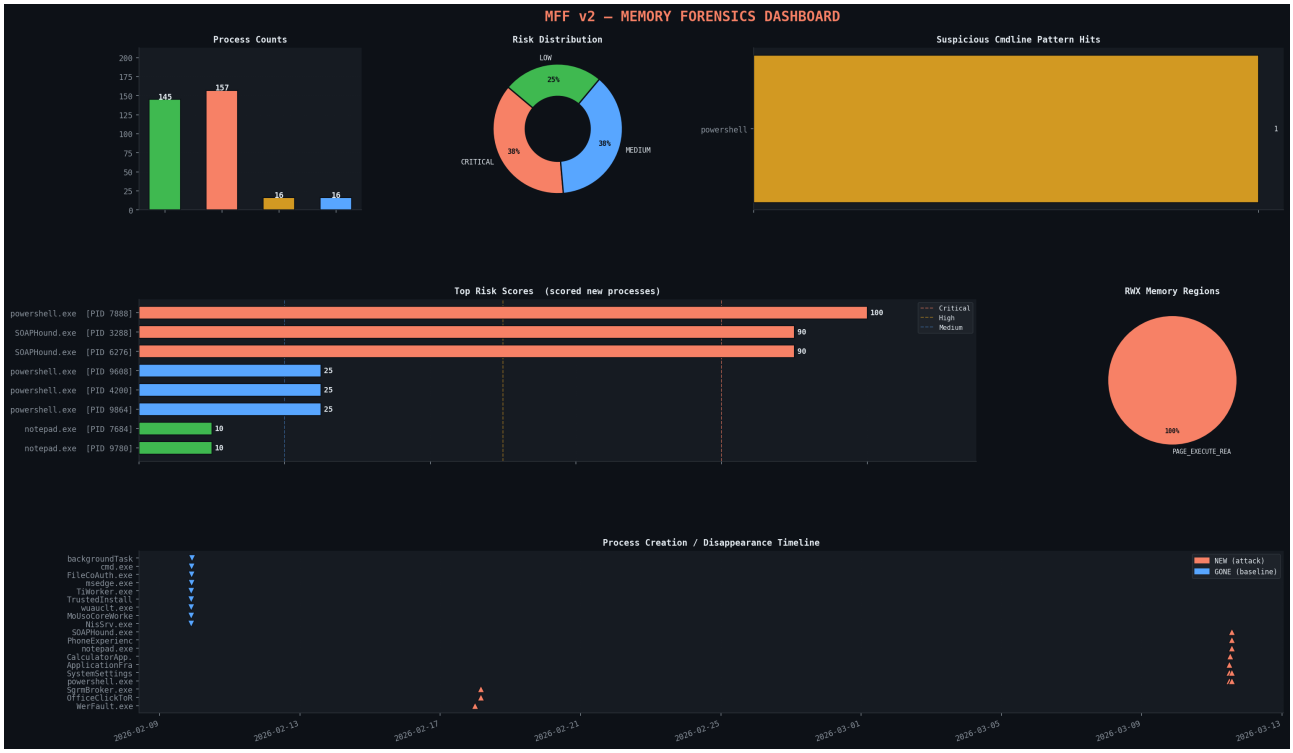
Defense Evasion

IOC Summary:

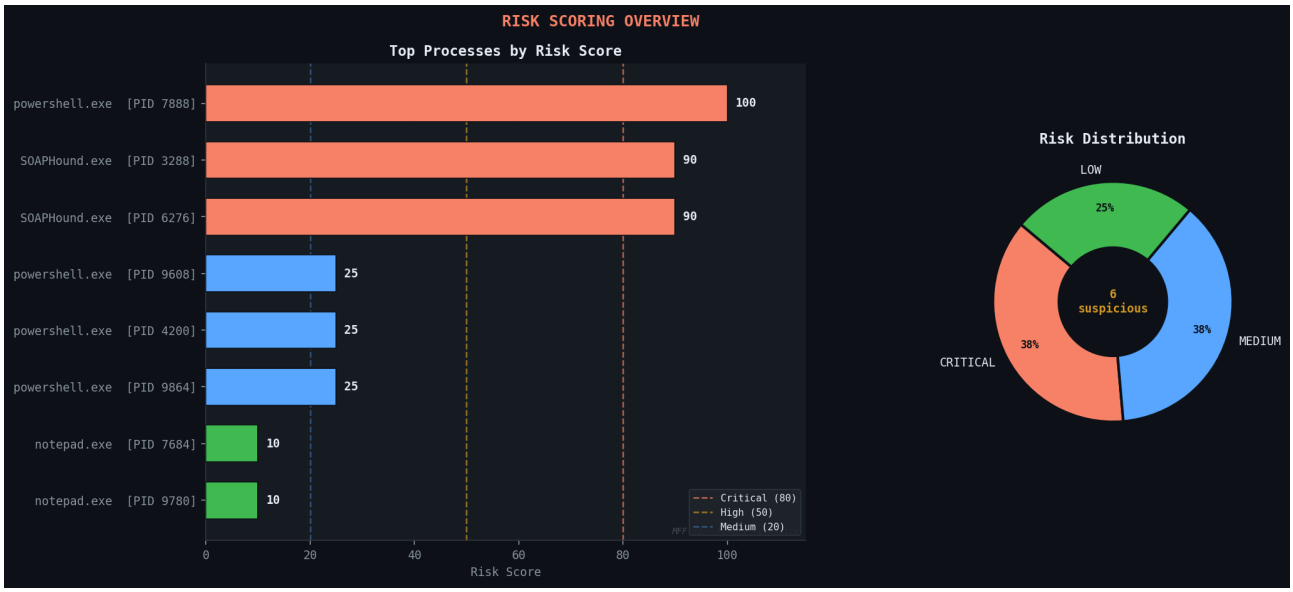
Type	Indicators (top 8)
IPv4	0.0.0.0 · 13.107.213.40 · 13.89.179.9 · 172.183.7.192 · 172.183.7.194 · 184.31.68.248

2. VISUALIZATIONS

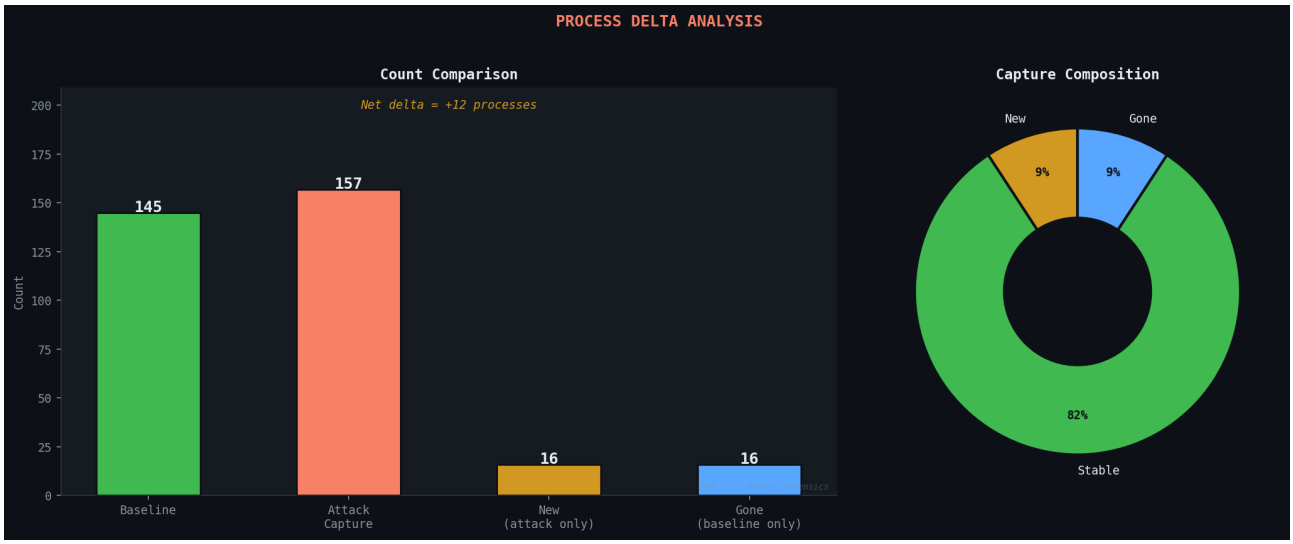
2.1 Full Forensics Dashboard



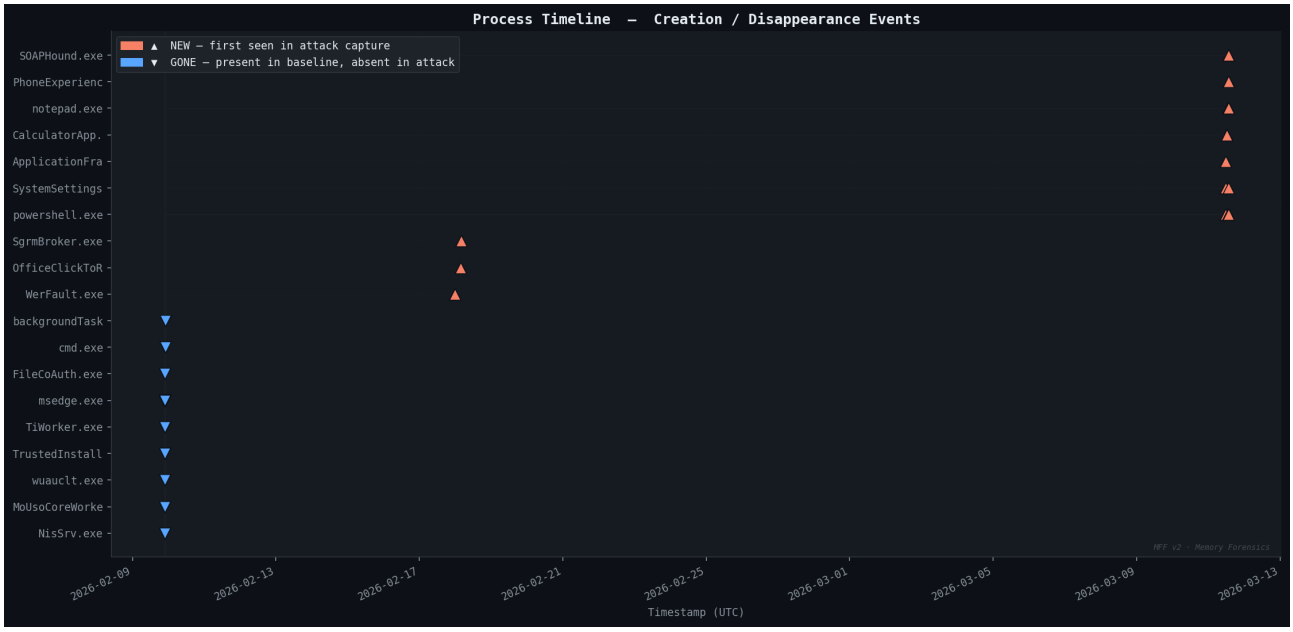
2.2 Risk Scoring Overview



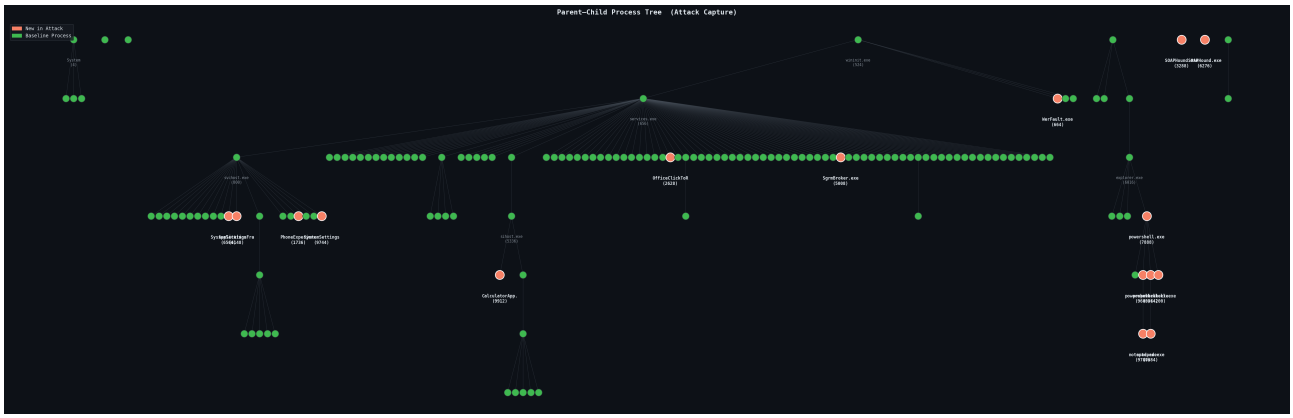
2.3 Process Delta Analysis



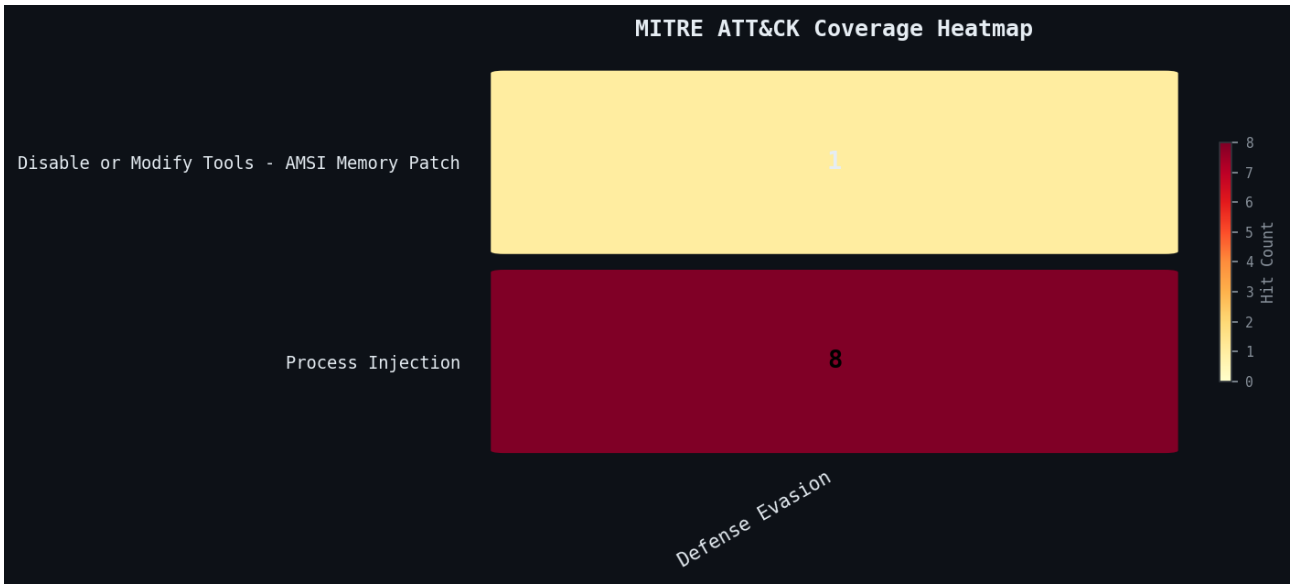
2.4 Process Timeline



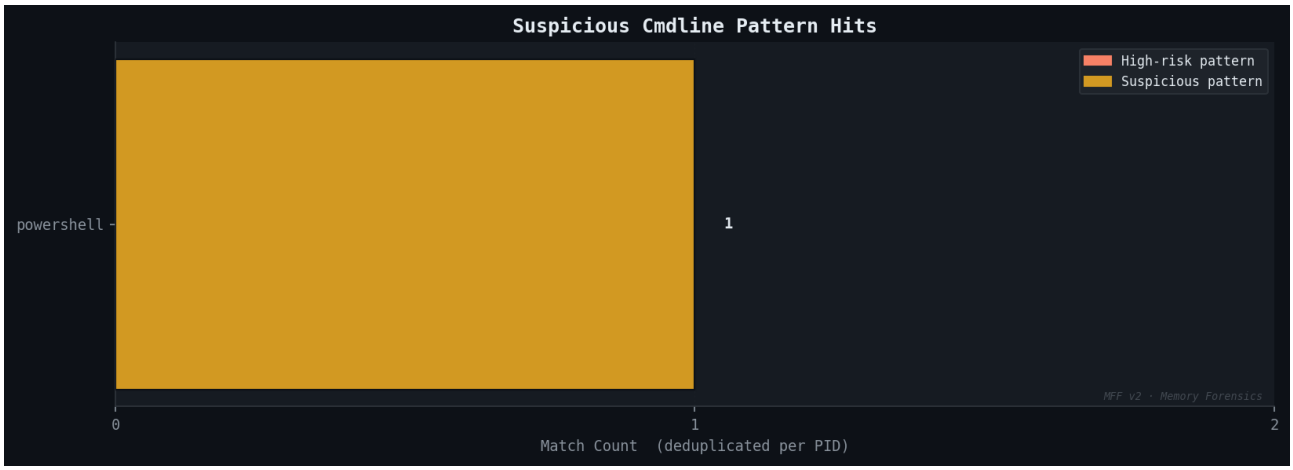
2.5 Parent-Child Process Tree


Parent-Child Process Tree (Attack Capture)
New in Attack
Baseline Process
System
(4)
wininit.exe
(544)
services.exe
(684)
svchost.exe
(800)
SgrmBroker.exe
(5408)
MsMpEng.exe
(4016)
powershell.exe
(7988)
CalculatorApp.
(8922)
SDXHelperBackgroundHost.exe
(5208)
(8176)
WerFault.exe
(964)
OfficeClickToR
(2628)
(5104)
(1216)
(9704)
(5016)
(7880)
(7880)
(7880

2.6 MITRE ATT&CK; Heatmap



2.7 Suspicious Cmdline Patterns



2.8 Malfind Memory Protections

3. RISK SCORE TABLE

Process	PID	Score	Level	Indicators		
powershell.exe	7888	100	CRITICAL	T1562.001	3 src	High
SOAPHound.exe	3288	90	CRITICAL	—	1 src	Medium
SOAPHound.exe	6276	90	CRITICAL	—	1 src	Medium
powershell.exe	9608	25	MEDIUM	—	1 src	Medium
powershell.exe	4200	25	MEDIUM	—	1 src	Medium
powershell.exe	9864	25	MEDIUM	—	1 src	Medium
notepad.exe	7684	10	LOW	—	1 src	Medium
notepad.exe	9780	10	LOW	—	1 src	Medium

4. NEW PROCESSES (Attack Only)

ImageFileName	PID	PPID	CreateTime
WerFault.exe	664	524	2026-02-17 23:49:52.000000 UTC
OfficeClickToR	2628	656	2026-02-18 03:45:58.000000 UTC
SgrmBroker.exe	5008	656	2026-02-18 03:48:05.000000 UTC
powershell.exe	7888	6016	2026-03-11 11:48:11.000000 UTC
SystemSettings	6564	800	2026-03-11 11:50:00.000000 UTC
ApplicationFra	6148	800	2026-03-11 11:50:18.000000 UTC
CalculatorApp.	9912	5336	2026-03-11 12:18:39.000000 UTC
powershell.exe	9608	7888	2026-03-11 13:18:21.000000 UTC
notepad.exe	9780	9608	2026-03-11 13:18:21.000000 UTC
powershell.exe	9864	7888	2026-03-11 13:24:48.000000 UTC
notepad.exe	7684	9864	2026-03-11 13:24:49.000000 UTC
PhoneExperienc	1736	800	2026-03-11 13:28:34.000000 UTC
powershell.exe	4200	7888	2026-03-11 13:29:21.000000 UTC
SOAPHound.exe	6276	9816	2026-03-11 13:32:02.000000 UTC
SOAPHound.exe	3288	9508	2026-03-11 13:32:03.000000 UTC
SystemSettings	9744	800	2026-03-11 13:32:54.000000 UTC

5. MITRE ATT&CK; TECHNIQUE MATCHES

Technique	TechniqueName	Tactic	MatchedKeyword	PID	Process
T1055	Process Injection	Defense Evasion	PAGE_EXECUTE_READWRITE	2856	MsMpEng.exe
T1055	Process Injection	Defense Evasion	PAGE_EXECUTE_READWRITE	5712	SearchApp.exe
T1055	Process Injection	Defense Evasion	PAGE_EXECUTE_READWRITE	7888	powershell.exe
T1055	Process Injection	Defense Evasion	PAGE_EXECUTE_READWRITE	1736	PhoneExperienc
T1055	Process Injection	Defense Evasion	EXECUTE_READWRITE	2856	MsMpEng.exe
T1055	Process Injection	Defense Evasion	EXECUTE_READWRITE	5712	SearchApp.exe
T1055	Process Injection	Defense Evasion	EXECUTE_READWRITE	7888	powershell.exe
T1055	Process Injection	Defense Evasion	EXECUTE_READWRITE	1736	PhoneExperienc
T1562.001	Disable or Modify Tools - AMSI Memory Patch	Defense Evasion	amsi.dll	7888	powershell.exe

6. SUSPICIOUS CMDLINE FINDINGS

ImageFileName	PID	MatchedPattern	Args
powershell.exe	7888	powershell	"C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe"

7. MALFIND — MEMORY INJECTION ARTEFACTS

Process	PID	Protection
powershell.exe	7888	PAGE_EXECUTE_READWRITE
powershell.exe	7888	PAGE_EXECUTE_READWRITE
powershell.exe	7888	PAGE_EXECUTE_READWRITE
powershell.exe	7888	PAGE_EXECUTE_READWRITE
powershell.exe	7888	PAGE_EXECUTE_READWRITE

7b. DLL ANALYSIS — T1574 / T1562 FINDINGS

1 DLL finding(s) — T1574 DLL Search Order Hijacking + T1562 AMSI Bypass evidence.

PID	Process	DLL	HijackType	Technique	RiskScore	Tactic
7888	powershell.exe	amsi.dll	AMSI_BYPASS_MEMORY_PATCH	T1562.001	80	Defense Evasion

[T1562.001] AMSI_BYPASS_MEMORY_PATCH — powershell.exe (PID=7888)

HIGH CONFIDENCE T1562.001: 'powershell.exe' (PID=7888) loads amsi.dll AND has 5 PAGE_EXECUTE_READWRITE region(s). Standard pattern for PowerShell AMSI bypass via memory patching.

8. NETWORK ARTEFACTS

8a. Flagged Connections (suspicious ports)

TreeDepth	Offset	Proto	LocalAddr	LocalPort	ForeignAddr	ForeignPort	State	PID	Owner	Created	ImageFileName	SuspiciousPort	PortMeaning
0	0xa4827a562e10	TCPv4	0.0.0.0	135	0.0.0.0	0	LISTENING	920	svchost.exe	2026-02-17 23:49:54.000000 UTC	svchost.exe	135	DCOM/RPC
0	0xa4827a563230	TCPv4	0.0.0.0	135	0.0.0.0	0	LISTENING	920	svchost.exe	2026-02-17 23:49:54.000000 UTC	svchost.exe	135	DCOM/RPC
0	0xa4827a563230	TCPv6	::	135	::	0	LISTENING	920	svchost.exe	2026-02-17 23:49:54.000000 UTC	svchost.exe	135	DCOM/RPC
0	0xa4827f8f3890	TCPv4	10.0.2.5	139	0.0.0.0	0	LISTENING	4	System	2026-03-11 13:27:02.000000 UTC	System	139	NetBIOS

9. IOC DETAIL TABLE

Type	Value	Source	Count
FilePath	C:\\Windows\\system32\\svchost.exe	cmdline	49
FilePath	C:\\Windows\\System32\\svchost.exe	cmdline	28
FilePath	C:\\Program	cmdline	12
FilePath	C:\\Users\\lucif\\AppData\\Local\\Packages\\Microsoft.MicrosoftOfficeHub_8wekyb3d8bbwe\\LocalState\\EBWebView	cmdline	7
FilePath	Microsoft\\EdgeWebView\\Application\\144.0.3719.115\\msedgewebview2.exe	cmdline	7
FilePath	C:\\Windows\\System32\\RuntimeBroker.exe	cmdline	6
FilePath	.....	malfind	4
FilePath	C:\\ProgramData\\Microsoft\\Windows	cmdline	2
FilePath	C:\\Windows\\system32\\DllHost.exe	cmdline	2
FilePath	Microsoft	cmdline	2
FilePath	Windows	cmdline	2
FilePath	system32\\csrss.exe	cmdline	2
FilePath	.....	malfind	1
FilePath	.....	malfind	1
FilePath	.....	malfind	1
FilePath	??\\C:\\Windows\\system32\\conhost.exe	cmdline	1
FilePath	C:\\Users\\lucif\\AppData\\Local\\Microsoft\\EdgeUpdate\\MicrosoftEdgeUpdate.exe	cmdline	1
FilePath	C:\\Users\\lucif\\AppData\\Local\\Packages\\Microsoft.MicrosoftOfficeHub_8wekyb3d8bbwe\\LocalState\\EBWebView\\Crashpad	cmdline	1
FilePath	C:\\Windows\\Explorer.EXE	cmdline	1
FilePath	C:\\Windows\\ImmersiveControlPanel\\SystemSettings.exe	cmdline	1
FilePath	C:\\Windows\\System32\\SecurityHealthSystray.exe	cmdline	1
FilePath	C:\\Windows\\System32\\SystemSettingsBroker.exe	cmdline	1
FilePath	C:\\Windows\\System32\\VBoxService.exe	cmdline	1
FilePath	C:\\Windows\\System32\\VBoxTray.exe	cmdline	1
FilePath	C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe	cmdline	1
FilePath	C:\\Windows\\System32\\oobe\\UserOOBEBroker.exe	cmdline	1
FilePath	C:\\Windows\\System32\\spoolsv.exe	cmdline	1
FilePath	C:\\Windows\\SystemApps\\Microsoft.Windows.Search_cw5n1h2txyewy\\SearchApp.exe	cmdline	1

Type	Value	Source	Count
FilePath	C:\\Windows\\SystemApps\\Microsoft.Windows.StartMenuExperienceHost_cw5n1h2txyewy\\StartMenuExperienceHost.exe	cmdline	1
FilePath	C:\\Windows\\SystemApps\\MicrosoftWindows.Client.CBS_cw5n1h2txyewy\\TextInputHost.exe	cmdline	1

10. METHODOLOGY

Acquisition: VBoxManage debugvm dumpvmcore — cold hypervisor-level acquisition. No kernel module or guest agent. Forensically sound. SHA256 + MD5 hash recorded on host immediately post-acquisition.

Volatility 3: Memory parsed using Volatility 3: pslist, pstree, cmdline, dlllist, malfind, netscan, threads. Exported as CSV (engine) and JSONL (audit trail).

Process diff: Baseline vs attack comparison by process name (ImageFileName). PID-based diff rejected — PIDs are reassigned on every reboot and produce false-positive churn exceeding 90% of all processes.

Risk scoring: Transparent rule-based scoring (no ML). RWX memory region +60 · Suspicious cmdline pattern +40 · Known-malicious process name +50 · Suspicious DLL load (T1574) +30 · Anomaly name bonus +15. Thresholds: CRITICAL ≥80 HIGH ≥50 MEDIUM ≥20 LOW <20.

Malfind filter: Known-clean JIT processes excluded (browsers, Defender, .NET, OneDrive). Self-match false positives suppressed (e.g. lsass.exe matching itself). Only genuinely suspicious PAGE_EXECUTE_READWRITE regions retained.

DLL analysis: windows.dlllist CSV analysed for T1574 DLL Search Order Hijacking. Protected DLLs (amsi.dll, version.dll, cryptbase.dll etc.) checked for loads from non-System32 paths. Findings merged into ATT&CK; tagged output.

ATT&CK; mapping: 30+ rule-based signatures across 8 MITRE ATT&CK; tactics. Keyword matching on cmdline args, process names, DLL paths, and network artefacts. T1574-specific patterns include: copy-item, amsi.dll, invoke-atomictest.

IOC extraction: Regex-based extraction: IPv4 (excluding RFC1918 private ranges), domains, URLs, MD5/SHA1/SHA256 hashes, Windows file paths.

Framework: MFF v2 — Python Post-Volatility Memory Forensics Framework. FYP project — University of Roehampton 2026. Tool chain: VBoxManage → Volatility 3 → MFF v2 → HTML + PDF reports. Modules: comparison_engine_v2, dll_analysis, mitre_tagger, network_ioc, process_tree, export_alert, report_generator.

Risk scoring: Transparent additive model — no ML. Each evidence category contributes once. Weights calibrated to specificity (ACPO P2). CRITICAL>=80, HIGH>=55, MEDIUM>=25.

Standards: ACPO Good Practice Guide for Digital Evidence (Principles 1-4). NIST SP 800-86 Integrating Forensic Techniques into Incident Response. ISO/IEC 27037:2012 Guidelines for Identification, Collection and Preservation. College of Policing Digital Forensics Guidance. CPS Disclosure Manual.